

Week 1: Strategic Planning and Data Exploration in Logistics

E-Commerce Order Fulfillment Analytics — Strategic Report

1. Project Definition

This project analyzes the E-Commerce Order Fulfillment Dataset (50,000 records) obtained from Kaggle. Each record represents one customer order and includes: Order_ID, Customer_Region (5 regions), Product_Category (6 categories), Order_Date, Ship_Date, Delivery_Date, Shipping_Mode (Standard/Express/Same Day), Shipping_Cost, Delivery_Status (Delivered/Delayed/Returned), and Delivery_Days. The scenario centers on order fulfillment reliability and cost: understanding how region, product category, and shipping mode relate to delivery speed, delay risk, and shipping cost.

Core business questions this project addresses:

- How much of total fulfillment time is warehouse processing (order-to-ship) versus carrier transit (ship-to-delivery)?
- Do faster/more expensive shipping modes actually reduce delay risk, or only cost more?
- Can delay or return outcomes be predicted from order attributes, and if not, what does that imply operationally?

Key Performance Indicators (KPIs)

Seven KPIs anchor the analysis, computed directly from the dataset (50,000 orders):

#	KPI	Definition	Result
1	Order Fulfillment Cycle Time	Average days from Order_Date to Delivery_Date (Order_to_Delivery_Days)	8.48 days
2	Ship Processing Time	Average days from Order_Date to Ship_Date (Order_to_Ship_Days) — isolates warehouse handling from carrier transit	2.49 days
3	Delivery Delay Rate	% of orders with Delivery_Status = 'Delayed'	15.01% (7,505 orders)
4	Return Rate	% of orders with Delivery_Status = 'Returned'	4.87% (2,433 orders)
5	Delivery Success Rate	% of orders with Delivery_Status = 'Delivered'	80.12% (40,062 orders)
6	Average Shipping Cost	Mean Shipping_Cost across all orders	\$138.84
7	Shipping Cost by Mode	Average Shipping_Cost split by Shipping_Mode	Standard \$99.29 / Express \$164.58 / Same Day \$299.59

2. Project Objectives

The main objectives of this project are:

1. To understand the order fulfillment process using the available e-commerce logistics data.
2. To measure delivery performance using key indicators such as average delivery time, delay rate, return rate, delivery success rate, and shipping cost.
3. To identify patterns in delivery performance and shipping cost across customer regions, product categories, and shipping modes.

4. To assess the quality and completeness of the available logistics data through missing-value, duplicate, data-type, and validity checks.
5. To establish a descriptive baseline that can be used for further logistics analysis and decision-making.
6. To develop a strategic roadmap for applying data science techniques such as classification, regression, and optimization in later phases of the project.
7. To use future predictive modeling to determine whether available order attributes can help predict delivery delays and delivery time.
8. To translate analytical findings into practical recommendations for improving delivery reliability and evaluating transportation cost efficiency.

3. Literature and Data Research

This project draws on four complementary data science technique families used widely in logistics analytics:

- Descriptive statistics: distributions and group-wise averages to establish baselines for delay rate, cycle time, and cost across categorical segments.
- Correlation analysis: examining linear relationships between numeric fields (cost, processing time, transit time) to detect which stages actually drive total cycle time.
- Classification: predicting a binary Delayed/Not-Delayed outcome from order attributes using models such as logistic regression or random forests.
- Regression: forecasting Delivery_Days as a continuous target to quantify expected transit time under different conditions.

Industry research generally finds that carrier and route account for most delivery-time variance, while order size effects are secondary — this project tests whether that pattern holds in the available fields, and reports the result honestly even where it may run against that expectation.

4. Strategic Roadmap

4.1 Data Collection

Load the 50,000-row CSV via pandas; confirm shape, dtypes, and value ranges for each column.

4.2 Data Cleaning

Audit for missing values, duplicate Order_IDs, invalid date ordering (Ship_Date before Order_Date, etc.), and statistical outliers in Shipping_Cost and Delivery_Days.

4.3 Exploratory Data Analysis

Profile delivery-day and cost distributions; break down delay rate and cost by region, category, and shipping mode; correlate numeric fields.

4.4 Predictive Modeling

In later phases, regression models will be developed to predict **Delivery_Days**, while classification models will be developed to predict **Delayed status** using suitable available features such as Customer_Region, Product_Category, Shipping_Mode, and Shipping_Cost.

The models will be trained and evaluated using appropriate data-splitting procedures and performance metrics.

4.5 Optimization & Recommendations

Translate EDA and modeling findings into concrete operational or cost recommendations (e.g., shipping-mode value assessment).

5. Code Illustration

5.1 Load and inspect

```
import pandas as pd
import numpy as np

df = pd.read_csv('E-Commerce_Order_Fulfillment_Dataset.csv')
print(df.shape)          # (50000, 10)
print(df.info())
print(df.isnull().sum())  # 0 missing in every column
print(df.duplicated().sum()) # 0 duplicate rows
```

5.2 Derive cycle-time fields

```
date_columns = ['Order_Date', 'Ship_Date', 'Delivery_Date']
for col in date_columns:
    df[col] = pd.to_datetime(df[col])

df['Order_to_Ship_Days'] = (df['Ship_Date'] - df['Order_Date']).dt.days
df['Order_to_Delivery_Days'] = (df['Delivery_Date'] - df['Order_Date']).dt.days
```

5.3 KPI calculation (all 7 KPIs)

```
# KPI 1: Order Fulfillment Cycle Time
avg_cycle_time = df['Order_to_Delivery_Days'].mean()

# KPI 2: Ship Processing Time
avg_ship_time = df['Order_to_Ship_Days'].mean()

# KPI 3: Delivery Delay Rate
delay_rate = (df['Delivery_Status'] == 'Delayed').mean() * 100

# KPI 4: Return Rate
return_rate = (df['Delivery_Status'] == 'Returned').mean() * 100

# KPI 5: Delivery Success Rate (use .mean(), not .sum(), to get a %)
delivery_success_rate = (df['Delivery_Status'] == 'Delivered').mean() * 100

# KPI 6: Average Shipping Cost
avg_shipping_cost = df['Shipping_Cost'].mean()

# KPI 7: Shipping Cost by Mode
cost_by_mode = df.groupby('Shipping_Mode')['Shipping_Cost'].mean()
```

5.4 Proposed Predictive Modeling Approach

The following workflow represents the proposed predictive modeling approach for later phases. It is included to illustrate the planned use of data science techniques and does not represent a completed predictive model in Week 1.

Classification — Predicting Delivery Delay

Select relevant order attributes

↓

Encode categorical variables
↓
Define target: Delayed / Not Delayed
↓
Split data into training and testing sets
↓
Train classification model
↓
Evaluate model performance
↓
Interpret important factors

Regression — Predicting Delivery Time

Select suitable features
↓
Define target: Delivery_Days
↓
Split data into training and testing sets
↓
Train regression model
↓
Evaluate prediction error
↓
Use results for logistics planning

6. Expected Contributions

The project is expected to provide the following outcomes:

- Establish a baseline for delivery performance, delays, returns, delivery success, and shipping cost.
- Identify patterns in delivery performance and shipping cost across customer regions, product categories, and shipping modes.
- Determine in later phases whether the available order attributes provide useful predictive information for delivery delays and delivery time.
- Evaluate the cost-service trade-off associated with different shipping modes.
- Develop data-driven recommendations for logistics planning and operational efficiency.
- Identify additional operational data requirements, such as carrier, warehouse, route, and promised-delivery information, where the current dataset is insufficient to explain the deeper causes of delays.

7. Conclusion and Expected Outcomes

This Week 1 strategic plan establishes a structured path from raw e-commerce order data to logistics insights.

The initial exploration provides baseline measures for delivery performance, delay rate, return rate, delivery success, and shipping cost. The strategic roadmap then extends from data validation and exploratory analysis to future predictive modeling and optimization.

In later phases, classification and regression models will be evaluated to determine whether the available order attributes can help predict delivery delays and delivery time. The resulting analytical evidence can support shipping-mode evaluation, operational improvements, and data-driven logistics recommendations.

Because the current dataset does not include important operational fields such as carrier, warehouse, route, or promised delivery time, additional operational data may be required in future phases to identify the deeper causes of delivery delays.

Overall, the Week 1 analysis establishes the foundation for a structured and data-driven approach to improving e-commerce logistics efficiency, delivery reliability, and transportation cost management.

8. References / Public Resources

- **Kaggle** — E-Commerce Order Fulfillment Dataset used for this project.
- **Python Pandas Documentation** — Used for data loading, cleaning, transformation, grouping, and descriptive analysis.
- **NumPy Documentation** — Used for numerical and statistical computation.
- **Scikit-learn Documentation** — Relevant for future classification, regression, preprocessing, and model evaluation.